

Optimization of Regularized Matrix Factorization for Movie Recommendation Systems

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Abstract

Recommender systems are a core component of modern digital platforms, enabling personalized content delivery to users. A fundamental challenge in such systems is the extreme sparsity of user–item rating matrices, which makes accurate prediction of unseen ratings difficult. In this project, we address the recommendation problem using regularized matrix factorization, framing it as a matrix completion and optimization task. Experiments conducted on the MovieLens Latest Small dataset show that stochastic optimization methods significantly outperform simple baseline predictors.

1 Introduction

User–item interaction data in recommendation systems is typically sparse, as users rate only a small subset of available items. Accurately predicting missing ratings is essential for generating high-quality personalized recommendations. Matrix Factorization (MF) is a widely used approach that models users and items in a shared low-dimensional latent space.

The objective of this project is to learn a low-rank approximation of the rating matrix that minimizes prediction error on unseen data, while controlling overfitting through regularization.

2 Problem Formulation

Let $R \in \mathbb{R}^{m \times n}$ denote the user–item rating matrix, where m is the number of users and n is the number of movies. The matrix is only partially observed; the set Ω contains the indices of known ratings. The task is to predict the missing entries of R .

We formulate the problem as the following optimization objective:

$$\min_{U, V} \sum_{(i,j) \in \Omega} (R_{ij} - u_i v_j^T)^2 + \lambda(\|u_i\|^2 + \|v_j\|^2) \quad (1)$$

where U and V are the latent user and item factor matrices, K is the latent dimension, and λ is the regularization parameter. In addition, global, user-specific, and item-specific bias terms are included to capture systematic rating behavior.

3 Methodology

We evaluate two optimization strategies:

- **Stochastic Gradient Descent (SGD):** Model parameters are updated iteratively using individual observed ratings, providing flexibility and fine-grained control over optimization.
- **Alternating Least Squares (ALS):** User and item factors are optimized alternately by solving least-squares subproblems.

Key hyperparameters include the latent dimension K , learning rate α , and regularization coefficient λ .

4 Experiments

4.1 Dataset and Setup

Experiments are conducted using the MovieLens Latest Small dataset, which contains approximately 100,000 ratings from around 600 users on nearly 9,000 movies. The dataset is split into 80

Baseline methods include a global mean predictor and a column-based (item-average) predictor.

4.2 Results and Analysis

Results show that SGD converges steadily across different values of the latent dimension K . Increasing K improves training accuracy but also increases the risk of overfitting. The observed generalization gap highlights the bias-variance trade-off inherent in model complexity.

Overall, matrix factorization optimized with SGD consistently outperforms baseline predictors and achieves substantially lower RMSE on the test set.

5 Conclusion

This project demonstrates that regularized matrix factorization is an effective approach for sparse recommendation problems. SGD provides strong empirical performance and flexibility, while ALS offers a useful point of comparison despite being less adaptable for extensive hyperparameter tuning.

6 Future Work

Future directions include incorporating implicit feedback, using adaptive learning rates, and developing hybrid recommendation models that combine collaborative and content-based information.